

TARGET TRACKING BY NEURAL NETWORK MANEUVER MODELING

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Abstract

A new approach to tracking a maneuvering target using a neural network-based scheme is developed. The neural network models the target maneuver and assists a Kalman filter in updating its gains in order to generate correct estimates of target position and velocity. A performance evaluation of the target tracking scheme is conducted under various interesting scenarios. The parallel processing capabilities of trained neural nets are exploited in this application for realistically handling more input features to correct for the bias induced by the target maneuver.

1 Introduction

Conventional target tracking algorithms based on statistical techniques perform quite efficiently when the target motion does not involve maneuvers. Target maneuvers involving short term accelerations, however, cause a bias in the measurement sequence, which unless compensated results in divergence of the Kalman filter that provides estimates of target position and velocity, in turn leading to a loss of track. Accurate compensation for the bias requires processing more samples of the input signals which adds to the computational complexity. The waiting time for more samples can also result in a total loss of track since the target can begin a new maneuver and if the target begins a new maneuver before the first one is compensated for, the filter would never converge. Most of the proposed algorithms in the current literature hence have the disadvantage of losing the target in short term accelerations, i.e., when the duration of acceleration is comparable to the time period between the measurements. The problem gets further compounded if target maneuvers in clutter environment are to be estimated.

In this paper we shall propose a neural network scheme for the modeling of target maneuvers. The primary motivation for employing a neural network for this task comes from the efficiency with which more features can be used as inputs for bias compensation. The parallel processing capability of a properly trained neural network can permit fast processing of features to yield correct acceleration estimates and hence can take the burden off the primary Kalman filter which still provides the target position and velocity estimates.

A principal feature of the present approach to tracking is the feasibility of using information in addition to the innovation sequence to facilitate maneuver detection and correction in one single step. Existing methods that attempt to use such additional information either require separate filters [1] or a higher dimensional state vector (augmented state) [2], both of which increase the computational complexity. While the innovation sequence is the one (and commonly used) parameter that provides an indication of maneuver, there are other available parameters each of which is capable of indicating a different characteristic of target maneuver. Classifying the type of maneuver performed by the target using more than one parameter is what makes the neural network more appealing for this task. The present scheme also enjoys a higher degree of fault tolerance and provides a more graceful degradation when a false maneuver is detected. This is due to the fact that the neural network recalculates its estimates at each sampling instant before the measurement bias builds up. It must be emphasized that detection and correction take place simultaneously after each sampling period, whereas in conventional techniques [3,4] these are two distinct phases unless more complex computations are allowed by available resources.

It must also be emphasized that the present neural network approach to target tracking is a novel one

and differs significantly (both in concept and application) from the recent work of Iltis [5, 6] that attempts to use a neural network in target tracking application. Specifically, the algorithms developed in [5, 6] focus on the data association problem in a multiple non-maneuvering (constant velocity) target environment, which is formulated as an optimization problem to which a neural network solution is proposed. In contrast, the present work focuses on the modeling of target maneuvers by a neural network which assists the Kalman filter in efficiently generating the estimates of the position and velocity of the maneuvering target.

This synergistic use of a neural network with an existing estimation scheme is what should make the present work of interest to a wider audience than the application to the target tracking problem would imply. Rather than discard an existing solution and propose a neural network as a total replacement (which may be unacceptable to a majority of conventional practitioners), the present approach attempts to tailor the existing features in the problem to a form that could be efficiently handled by a neural network which is used as a component assisting the main filter in the overall solution. It is hence believed that this work would be of interest to the general neural network community besides the estimation and filtering communities.

2 Neural network architecture and training details

For the sake of precision, the development in this section and the performance evaluation in the following section will be given for the neural network modeling of longitudinal acceleration of the target. The present approach can be readily extended to track other types of maneuvers (e.g., circular) also. We will employ a three layer feedforward network trained by error backpropagation for obtaining as outputs the estimates of the two components of target acceleration.

Of fundamental importance for a satisfactory training of the neural network is the identification of the set of input features to be used. The performance evaluations that will be discussed in the next section were conducted with the use of three input parameters which will be briefly outlined in the following. It may be noted that the three specific quantities that are useful in obtaining an accurate estimate of target acceleration are (i) intensity of acceleration, (ii) direction of tangential velocity, and (iii) initial velocity at

the time of acceleration. The three input parameters selected are designed to provide this information to the neural network.

The first input parameter is the innovation term multiplied by the Kalman filter gain to smooth out the predicted estimate, i.e.,

$$\nu(k) = z(k) - H\hat{x}(k|k-1)$$

where $z(k)$ denotes the measurement at instant k and $\hat{x}(k|k-1)$ is the estimate at instant k based on the past measurements upto the $(k-1)^{th}$ instant. This expression is representative of the actual "newness" of information about position measurements. The change in residual information is sensitive to changes in velocity and therefore different initial velocities are included in the training set. In terms of the duration of acceleration, however, each estimate is updated after a sampling instant. The changes in position innovation are normalized with respect to the covariance of the innovation $S(k)$.

The second input parameter is the heading estimate. The estimation of heading from noisy position measurements is by itself a problem of considerable practical importance. The common approach to this problem is to measure x and y positions simultaneously and assuming a constant velocity, the heading estimate is obtained as $\hat{y}/\hat{x}$ where the velocity estimates $\hat{x}$ and $\hat{y}$ are derived from a least square estimator. However, the assumption of constant velocity is not valid in our problem and hence the estimate is obtained as a line fit through all the measurements. The method of least triangles [7] is used to compute the heading estimate with the standard deviations σ_x^2 and σ_y^2 being assumed unknown.

For providing a measure of the target velocity at the start of the maneuver, we use the third input parameter. The velocity estimate from the Kalman filter output cannot be used for this purpose since it is constructed from a number of samples after the maneuver has begun and hence is biased information. Consequently, assuming that doppler measurements are available, we use the increment in the normalized doppler shifts

$$\Delta f_d(k) = (f_d(k)/\sigma_{f_d})^2 - (f_d(k-1)/\sigma_{f_d})^2$$

where $f_d(k)$ and $f_d(k-1)$ are the doppler shifts at instants k and $k-1$ respectively, and $\sigma_{f_d}^2$ is their variance.

The maneuver modeling performance of a neural network with one hidden layer comprising of 14 nodes with nonlinear activation functions $f(z) = (1+e^{-z})^{-1}$

will be described in the following section. The two output nodes are selected as linear nodes. The network was trained with 800 training vectors in the batch processing mode using generalized delta rule with momentum [8] for adjusting the weights and with the learning rate selections .003 for the hidden layer and .018 for the output layer.

For generating the training data, a series of examples where a longitudinal acceleration was performed at different initial velocities were considered, with the expected range of target velocity 200m/sec to 700m/sec. The expected range of acceleration was taken to be 0 – 20m/sec² and training examples were generated at 1m/sec² increments. The examples are based on the maximum tolerable tracking error per radar scan, which corresponds to a 10m/sec error in speed (which is typical of an error induced by clutter variations). To prepare for the worst case, we assume a doppler uncertainty of 30m/sec in the velocity innovation, which results in 15% error for the minimum velocity. Several straight trajectories along which target accelerations were executed were considered with heading angles in the range [0 – 45°].

3 Target tracking performance

For evaluating the target tracking performance of the present neural network scheme and for comparing it with the performance delivered by the existing methods, several simulation exercises that included different target maneuvers have been conducted. These have indicated a consistently superior performance in tracking short term accelerations by the three layer neural network trained in the last section. Due to page restrictions, the highlights from only a few of these experiments will be briefly outlined.

In the first experiment, a maneuver is executed by a target which follows a straight trajectory from the initial position of (100m, 100m) with respect to the radar with an initial speed of 250m/sec and is radially moving away from the radar with a heading of 45°. The scan period T was assumed to be 10 seconds. The standard deviation of the measurement error σ_r was assumed to be dependent on the range with a maximum of 100 meters. Therefore, as the target moves away from the radar, σ_r increases to a maximum of 100 meters. The azimuth standard deviation σ_θ was taken to be .003 radian. The maneuver consists of a sharp acceleration of 5m/sec² performed at $t = 40$ sec. and lasting for only 10 seconds (which is one sampling period). The mean filtering errors

for the x -coordinate velocity and the y -coordinate position (selected for illustration) are shown in Figs. 1a and 1b, where the performance of the neural network (NN) scheme is compared with the Input Estimation (IE) scheme due to Bogler [9] which is one of the best available methods for tracking maneuvering targets. The significant improvement in performance delivered by the NN scheme is clearly evident.

The performance improvement is even more dramatic when the target executes a combination of maneuvers. In this experiment, the target trajectory is maintained the same as before with the target initial position at (100m, 100m) and the initial velocity of 200m/sec. The first maneuver takes place at $t = 60$ sec. with an acceleration of 5m/sec² lasting 10 seconds, which is followed by a second maneuver occurring at $t = 80$ sec. (one sampling period after the first maneuver ends) with an acceleration of 20m/sec² lasting 20 seconds. The measurement uncertainty was retained at the same level as in Experiment 1. The x -coordinate position error plots for the NN and IE scheme are compared in Fig. 2. The overall performance is also compared in the table below, which indicates a 100% increase in the mean track life for the NN scheme.

Table 1			
Method	% Detection of target along the path	% Detection out of correlation gate	Mean track life in # of scans
NN	93.2	2.6	16/20
IE	82.4	10.0	8/20

A comparison of the performance of the neural network modeling scheme was also made with that of a tracking algorithm that permits incorporating radial velocity measurements. The classical Singer model [10] which is based on a correlated noise process that can be represented by an autoregressive (AR) process driven by white noise provides the best performance under these conditions. The radial velocity information speeds up the initialization phase because it requires only one sample to indicate the target speed instead of two or more samples needed by the position measurements. However, when the target path deviates from the radial approach to the radar, the radial velocity measurement gives only a poor approximation to the actual target velocity and contributes to tracking errors. With the neural network model, however, the target heading information is used as an additional input to train the network. Thus the relevance of doppler information to actual target speed is utilized in the training process and hence the performance does not degrade severely as the target approaches the radar in a nonradial path.

For obtaining a precise quantitative comparison, the following simulation is conducted. A target motion is started from initial position (10km, 10km) with initial velocity of 350m/sec in a radial trajectory corresponding to the heading angle $\alpha = 45^\circ$. Radar scan period is assumed to be 5 seconds. A maneuver is started at $t = 75\text{sec}$. which involves a longitudinal acceleration of 20m/sec² lasting 150 seconds. For the Singer model, the size of the correlation gate is set to 100. Under these conditions, the tracking performance delivered by the neural network scheme is comparable to that of the Singer model as depicted in Fig.3a which plots the mean position error for the two schemes. However, for a nonradial trajectory motion (corresponding to $\alpha = 20^\circ$), the performance from the Singer model degrades considerably whereas the neural network model maintains the same level of performance as before. The mean position errors for this case are sketched in Fig.3b. This comparison clearly indicates the benefits of including the heading information input which can be realistically handled due to the parallel processing abilities of the neural network.

4 Conclusion

The synergistic functioning of a trained neural network with a Kalman filter that provides estimates of the position and velocity of a maneuvering target is the principal feature of the present approach. The feasibility of employing the neural net for maneuver modeling and for updating the Kalman filter gains in order to correct for the bias induced by target maneuvers is demonstrated through experiments depicting several maneuver scenarios. While the performance delivered by the present scheme is shown to exceed that possible by existing approaches in these scenarios, additional performance evaluation in several other scenarios (tracking in clutter environment, for instance) attest to the strength of this approach (due to page restrictions, these results are not included here but will be presented at the conference). This work hence makes a useful contribution of the application of neural network technology in the filtering and estimation areas.

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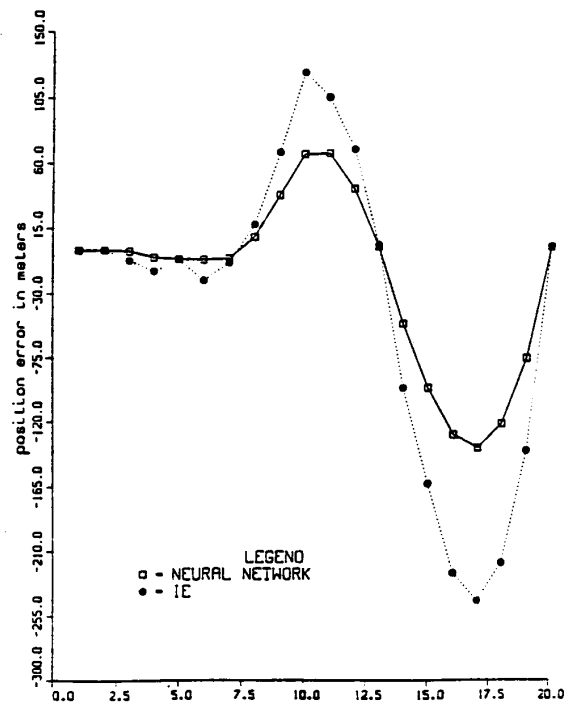


Fig-1a Comparison of NN and IE x co-ordinate velocity error.

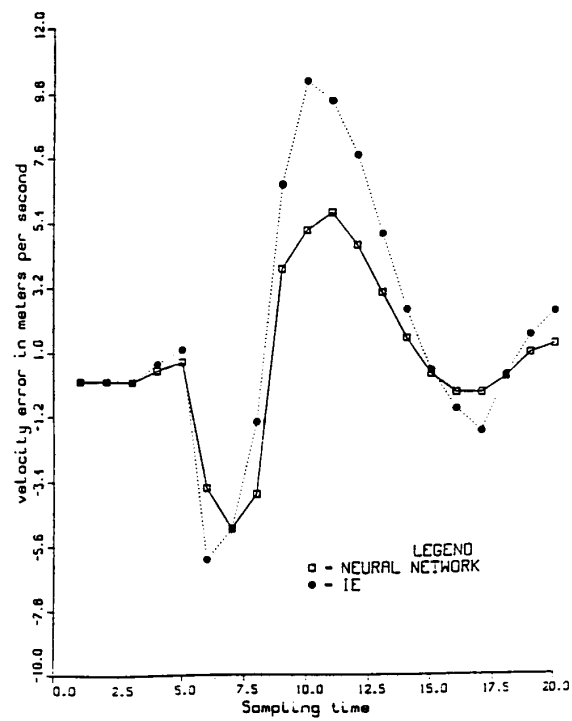


Fig-1b Comparison of NN and IE y co-ordinate position errors.

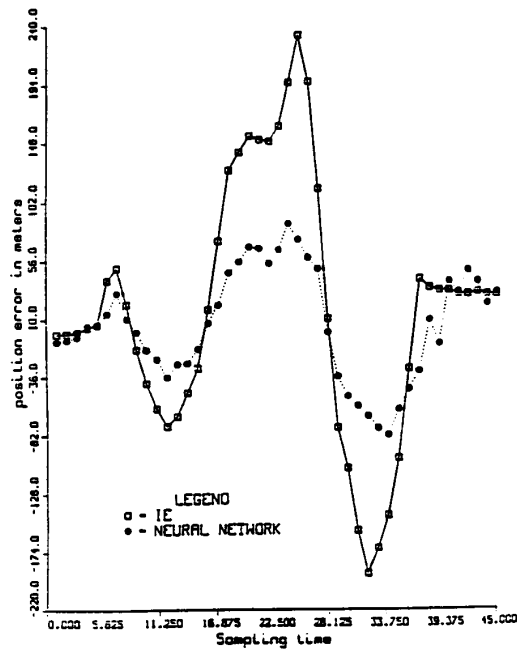


Fig-2 Comparison of the Mean filtering errors for NN and IE.

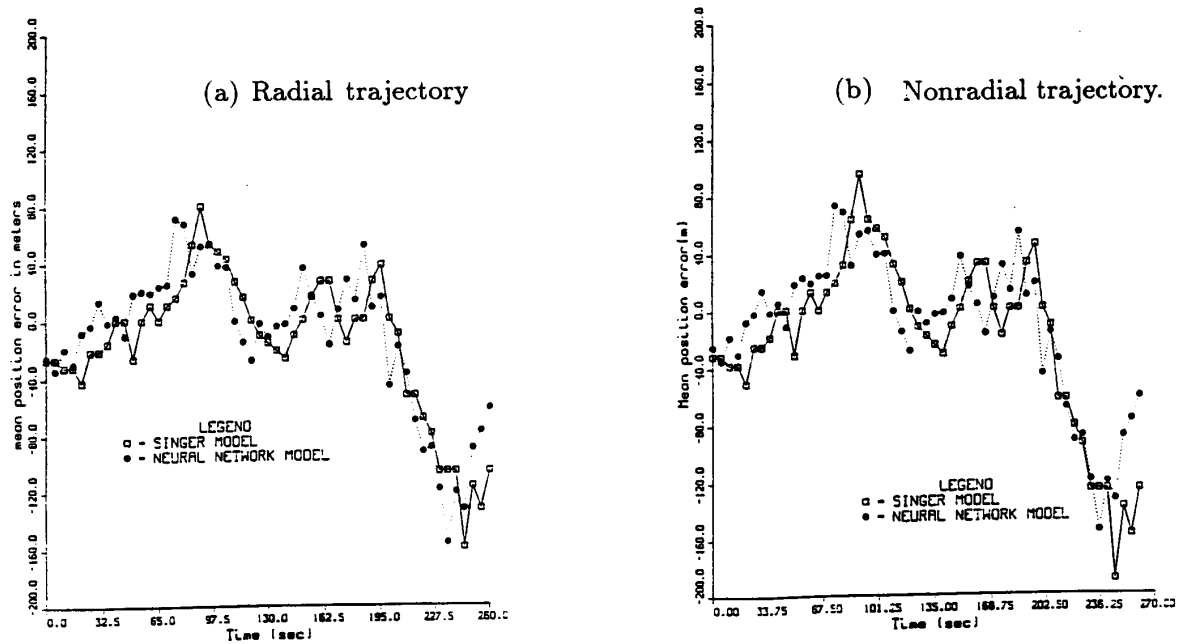


Fig-3 Comparison of NN scheme with Singer Model